

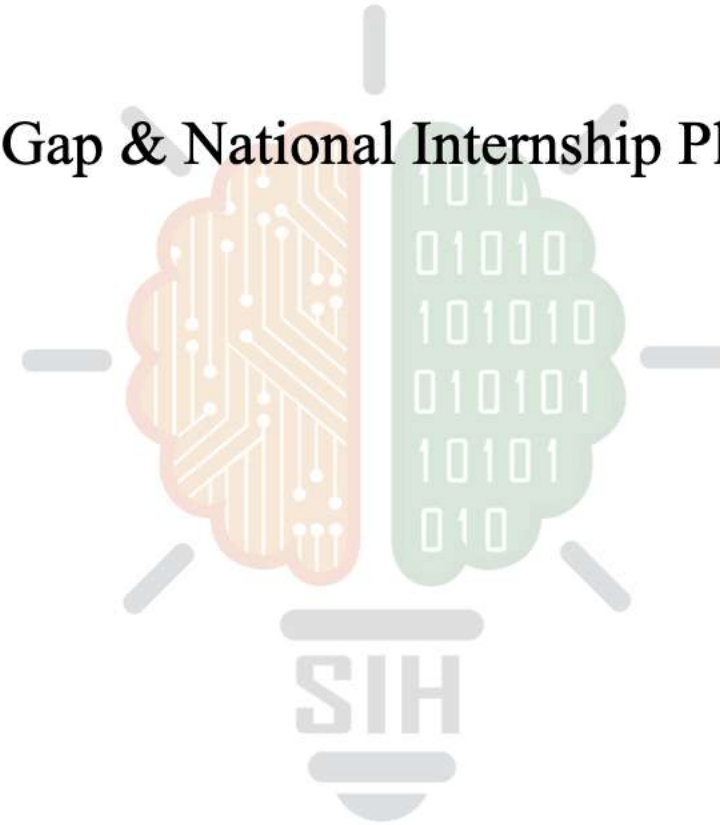
SMART INDIA HACKATHON

2026 TITLE PAGE



SMART INDIA
HACKATHON
2025

- **Problem Statement ID** - 25081
- **Problem Statement Title** - Academia–Industry Skill Gap & National Internship Platform
- **Theme** - Smart Education
- **PS Category** - Software
- **Team ID** - 00000
- **Team Name** – AyurSetu



Problem:

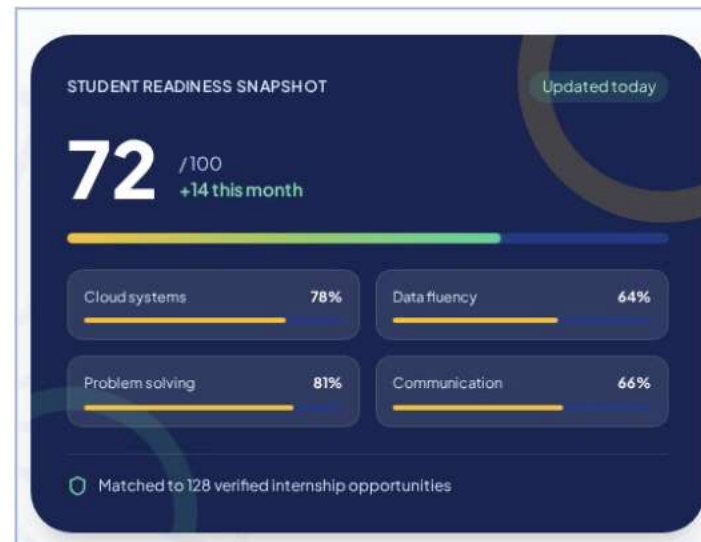
Students guess what skills to learn without standardized industry benchmarks. **Recruiters** spend months filtering thousands of resumes with **unverified claims**. Colleges teach **static syllabi** with no visibility into fast-changing enterprise frameworks.

Our Idea :

AyurSetu is a national **digital public infrastructure** that connects student capability, industry demand and academic action in **one trusted workflow**. It benchmarks a student against live enterprise hiring criteria, then **closes the gap and proves it** to employers.

Proposed Solution :

- **Skill Assessment Engine** evaluates real-world technical scenarios and auto-generates a **6-axis radar gap profile**.
- **Gap-closing roadmap** of AICTE-certified bridge courses mapped to each measured delta.
- **National Internship Board** ranks openings by verified competency with **1-click apply**.
- **Verified digital portfolio** — employer-readable, tamper-evident credentials.



Live student readiness snapshot — working prototype

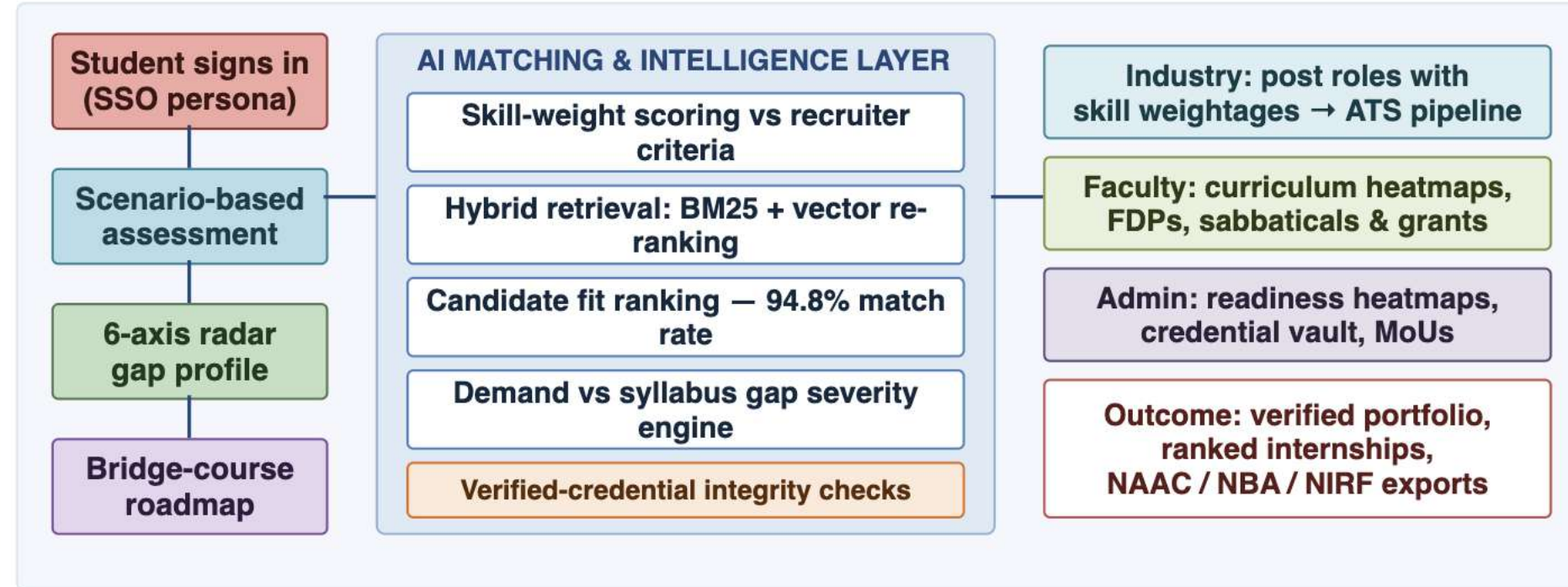
Innovation/Uniqueness:

- **India's first tripartite network** — students, faculty, industry and institution admins on one platform.
- **AI Candidate Fit Ranking** at **94.8% match rate** replaces keyword resume filtering.
- **Curriculum Modernization Engine** reads **10,000+ live postings** to flag obsolete syllabus modules (NEP 2020).
- **Credential Vault** auto-issues NOCs and NAAC / NBA / NIRF compliance exports.

Technology Stack

- **React 18 + TypeScript + Vite:** multi-persona SPA for student, faculty, industry and admin hubs.
- **Tailwind CSS:** government-grade accessible design system.
- **Recharts:** 6-axis radar competency maps, readiness heatmaps and placement trend analytics.
- **Skill-weighted matching engine:** scores candidates against per-skill recruiter weightages.
- **Hybrid retrieval (BM25 + vector):** maps postings to syllabus modules for gap detection.
- **Credential vault:** hash-verified NOCs, certificates and QR-stamped exports.
- **Deployed on Vercel:** live public prototype, SSO persona routing.

FLOW CHART



FOUR-PERSONA ARCHITECTURE



Live prototype:
ayursetu-five.vercel.app
Judge access: 1-click persona switch

All 4 stakeholder hubs are functional

Feasibility

- **Working prototype already live** and publicly deployed — all four stakeholder hubs functional, no hardware dependency.
- **Pure software, web-first:** runs on any browser; zero device cost to a student in a Tier-3 college.
- **Institutional pathway exists:** designed as an **MoE & AICTE** initiative, so adoption rides on existing university compliance workflows (NAAC, NBA, NIRF).
- **Data already modelled:** 6 engineering departments, 3,420-student cohort, and a 6-month placement trajectory are demonstrated end-to-end.
- **Persona SSO** lets a judge or a Dean inspect any role in one click.

Challenges

- **Credential trust:** self-claimed skills are worthless to a recruiter unless independently verified.
- **Industry benchmarks shift fast** — a static question bank is obsolete within two hiring cycles.
- **Faculty bandwidth:** syllabus revision needs academic senate approval, not just a dashboard.
- **Cold-start:** the network is only valuable once both students and recruiters are present.

Viability & Validation

- **Demand validated in-platform:** assessments benchmarked against **5,000+ active internship requirements** and **10,000+ national tech postings**.
- **Matching accuracy:** AI candidate fit ranking demonstrates a **94.8% match rate**, cutting recruiter shortlisting to a pre-ranked pool.
- **Outcome proven on cohort data:** **892 internships secured** against a target of 800 — **+11.5% ahead**.
- **Sustainable model:** government-funded core utility; enterprises fund bootcamps, sponsored labs and R&D grants (**₹45 lakh** active in the faculty module).

Strategy to Overcome

- **Tamper-evident credential vault** with verification hashes and QR-stamped exports; NOCs issued by the Dean, completion letters by the employer.
- **Benchmarks recomputed from live postings** so question weightages track real hiring criteria automatically.
- **Gap severity is delivered as an actionable revision note** plus a ready-to-teach FDP, not raw analytics.
- **Seed via institutions, not individuals:** onboard a university's full cohort at once, with MoU-signed partners on the demand side.

Direct Impact on Target Users

- **Students:** replaces guesswork with a measured 6-axis gap profile. Critical, because only **4.4%** of engineering students can write a fully working program though **92%** can explain the concept.
- **Recruiters:** receive a **pre-ranked** candidate pool instead of unverified resumes — **340+** applicants auto-filtered at **94.8%** match rate in the prototype.
- **Faculty:** demand heatmaps convert into syllabus revisions and funded FDPs — addressing the AICTE mandate for **annual industry-consulted curriculum revision**.
- **Institutions:** one-click **NAAC / NBA / NIRF** compliance exports and a verified credential vault.

Strategic Impact

- Directly serves the **AICTE mandate of three internships** (4–8 weeks each) before graduation by making discovery and NOC issue a single workflow.
- Operationalises **NEP 2020** experiential learning with auditable skill evidence rather than attendance records.
- Levels the field for **Tier-2 and Tier-3** candidates, where IT hiring grew **over 50%** in H1 2025 versus 12–15% in Bengaluru and NCR.

Economic & Strategic Benefits

- **Addressable scale: 15.98 lakh** B.Tech seats were approved for 2025–26 across **5,875** AICTE-approved colleges — every one of them a potential deployment.
- **Closes a measured national gap:** engineering employability slipped to **70.15%** in 2026 from 71.5% in 2025, and the Mercer-Mettl index puts overall graduate employability at just **42.6%**.
- **Targets the scarcest skills:** AI/ML postings grew **over 600%**, yet only **46%** of graduates are proficient — exactly the deltas our bridge roadmap closes.
- **Fits a narrowing funnel:** India's top five IT firms plan **~82,000** fresher hires in FY2026, so competency proof, not volume, decides placement.



Gap & Problem Identification

- India's engineering employability fell to **70.15%** in 2026 from **71.5%** in 2025; the Mercer-Mettl index (10 lakh+ learners, 2,700 campuses) puts overall graduate employability at **42.6%**, down from 44.3% in 2023. *[India Skills Report 2026; Mercer-Mettl GSI 2025]*
- Knowing ≠ doing:** **92%** of students explain concepts correctly in MCQ form, **38%** partially write the code, only **4.4%** produce a working efficient program. *[Aspiring Minds benchmarking study]*

Technology Benchmarking

- Most in-demand skills are **AI & data analytics, cloud and cybersecurity**; AI/ML postings grew **over 600%** while only **46%** of graduates are rated proficient. *[India Skills Report 2026]*
- 70%** of IT firms and **50%** of BFSI firms already use AI in recruitment — validating an AI fit-ranking layer as the industry norm, not a novelty. *[India Skills Report 2026]*

Policy & Ecosystem Analysis

- AICTE mandates **three internships of 4–8 weeks** before graduation, **annual industry-consulted curriculum revision**, NBA accreditation and annual faculty refresher courses. *[PIB, Ministry of Education — Initiatives of AICTE]*
- NEP 2020** promotes experiential, multidisciplinary learning; Union Budget 2025–26 earmarked **₹2,000 crore** for AI infrastructure and research. *[ET Education, Mar 2026]*

Economic & Strategic Landscape

- Indian IT is projected at **\$315 bn** revenue in FY26 (+6.1% YoY) adding **~1,35,000** net new jobs, but the top five firms plan only **~82,000** fresher hires. *[NASSCOM Strategic Review 2026]*
- GCCs** added 6 lakh+ jobs (2019–24), projected **2.8–4 million by 2030**, demanding specialist depth over general software skills. *[NASSCOM; ET Education]*

Supply-Side & Market Sizing

- 15.98 lakh** B.Tech seats approved for 2025–26 (+7% YoY) across **5,875** AICTE-approved colleges; enrolment hit **12.53 lakh**, an eight-year high. *[AICTE Annual Report 2024–25]*
- Structured hiring cannot absorb 12–15 lakh graduates a year — confirming the bottleneck is **alignment**, not headcount. *[ET Education analysis, Mar 2026]*

Prototype Results & Validation

- Live deployment benchmarks against **5,000+** internship requirements and **10,000+** national postings: **84.6%** readiness, **892** internships secured vs 800 target, **78** MoU partners. *[AyurSetu platform, ayursetu-five.vercel.app]*
- Department heatmaps span 6 branches (CSE 88% to Civil 69%) over a **3,420**-student cohort, with demand concentrated in cloud (34%) and AI (28%). *[AyurSetu platform data]*